ReflectAI

AI-Powered Retroreflectivity Measurement System

Team CUTIEFR
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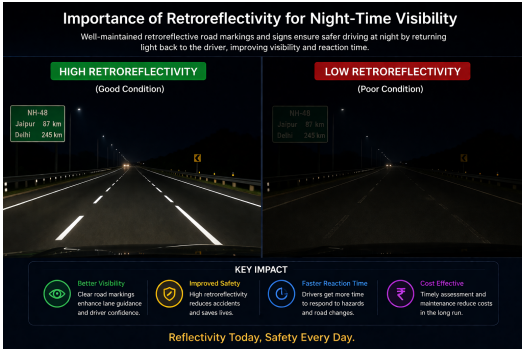
The Problem: Night-Time Highway Safety

The Challenge

- **73%** of highway fatalities occur at night.
- **1.4 lakh+ km** of NH require monitoring.
- Road markings lose $> 50\%$ reflectivity in 2 years.

The Manual Limitation

- **Slow:** Only 5 km/day.
- **Unsafe:** Technicians on live highways.
- **Expensive:** Rs. 8,000+/km.



Comparison: High vs Low Retroreflectivity

Our Solution: ReflectAI

Core Innovation: A vehicle-mounted AI system that converts pixel brightness into RA coefficients.

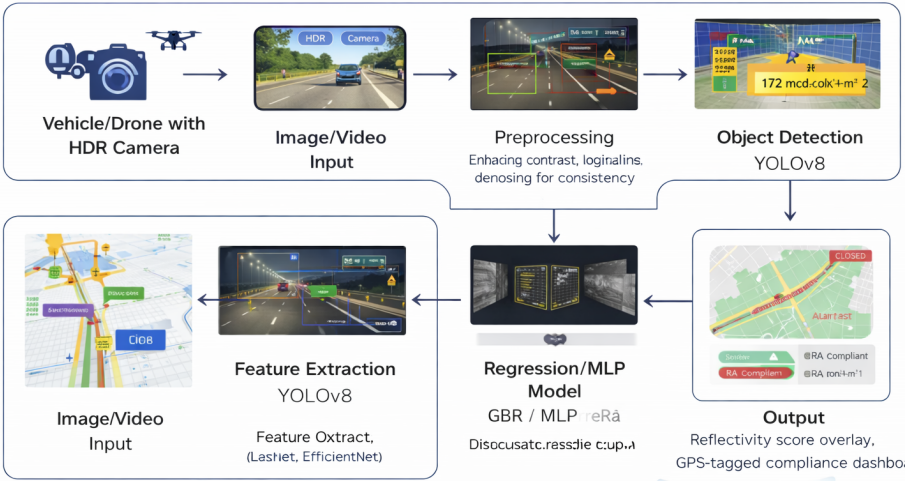
- **YOLOv8:** Real-time detection of signs and markings.
- **EfficientNet-B4:** Predicts reflectivity from imagery.
- **Groq LLaMA:** Auto-generates official work orders.

Efficiency Gain

100× Faster | 40× Cheaper than Manual Methods

Rs. 200/km vs Rs. 8,000/km

System Architecture – 7-Stage Pipeline



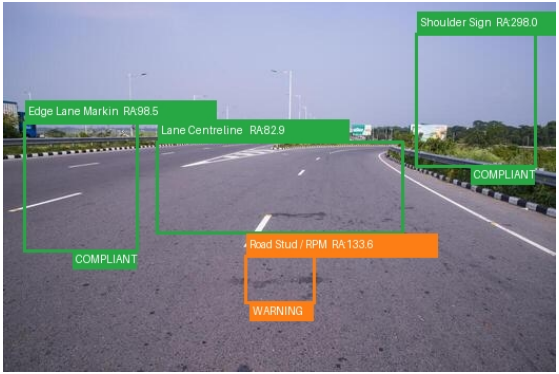
AI/ML Pipeline Details

Detection (YOLOv8)

- Detects markings, signs, and studs.
- Target mAP@0.5 > 0.85.
- 80+ FPS on standard CPU.

RA Prediction

- EfficientNet-B4 features.
- Gradient Boosting Regressor.
- **IRC 67 & 35** Compliance mapping.



Groq AI Maintenance Agent

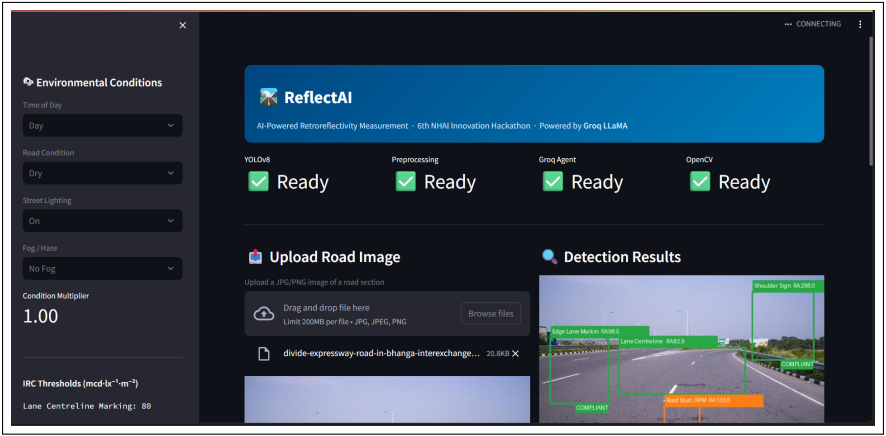
Agentic Workflow (LLaMA 3.3-70B):

1. `analyze_scan_results`
2. `prioritize_maintenance`
3. `estimate_repair_cost`
4. `generate_work_order`

Live Demo Performance

- Work order generation: <**5 seconds**.
- Automatic Tier 1/2/3 prioritization.
- REST API ready for NHAI AMS.

Live Demo Results – Streamlit Interface



- **Green/Orange/Red** status based on IRC thresholds (e.g., 80 mcd for markings).
- Real-time condition multiplier (Day/Night/Fog).

Performance & Scalability

Operational Metrics

- **Coverage:** 600 km/day.
- **Latency:** <2 sec/frame.
- **Hardware:** NVIDIA Jetson Nano + 4K Camera.

Scalability

- No specialized sensors required.
- Scalable across the entire 50,000+ km NHAI network.
- Open-source implementation (Python/PyTorch).

Conclusion – Why ReflectAI?

- **Fully Automated:** Removes human error and risk.
- **Cost Efficient:** 40× cheaper than current standards.
- **IRC Compliant:** Pre-configured for Indian Highway standards.
- **Ready for Deployment:** Scalable, cloud-integrated, and intelligent.

Reflectivity Today, Safety Every Day.

Thank You

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